

HeartCompass Project Charter

Course Project Charter for the HeartCompass platform

Project Name	HeartCompass
Project Period	March 30, 2026 to June 14, 2026
Project Type	AI software course project
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Estimated Workload	624 hours
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Project Overview

HeartCompass is planned as a dual-mainline AI platform that will transform fragmented interpersonal information into structured relational intelligence and an interactive virtual figure experience. The course project will deliver a lightweight client, a context-aware analysis engine, and a memory-grounded virtual figure workflow within a single integrated schedule.

Primary Goal

The project will build a platform that collects user and relationship context, organizes it into retrievable business entities, and uses AI to support both communication analysis and virtual figure interaction.

Objectives

The project will pursue four closely related objectives. First, it will collect and structure user, crush, and relationship data so that the system has a dependable foundation for reasoning. Second, it will transform narratives, screenshots, and curated knowledge into retrievable contextual assets. Third, it will generate communication guidance that helps users interpret social situations and choose appropriate next actions. Fourth, it will support a virtual figure workflow that simulates context-grounded dialogue on the basis of persona, relationship history, and recalled memory.

Success Criteria

The project will be considered successful if it can organize fragmented interpersonal materials into stable business entities, produce communication guidance that is specific and actionable, and support a virtual figure whose replies remain consistent with stored persona and relationship context. Success will also require the integrated client and backend to demonstrate the complete workflow within the planned schedule and estimated workload.

Scope Anchors

The planned system scope will cover the public capabilities exposed through `/addKnowledge`, `/addContextByNaturalLanguage`, `/addContextByScreenshots`, `/recallContextFromEmbedding`,

/conversationAnalysis, /narrativeAnalysis, /recalculateContextBlock, WebSocket /virtual_figure/send. At the business level, the platform will rely on the core entities User, Crush, RelationChain, Event, ChatTopic, Knowledge, InteractionSignal, DerivedInsight, ContextEmbedding. These interfaces and entities define the functional boundary of the course project and will be treated as the main scope anchors in later planning, review, and acceptance.

Assumptions and Risks

The charter assumes that users will provide enough narrative, profile, or screenshot information for the system to construct meaningful context, and it also assumes that MBTI-based knowledge and relationship history will improve grounding quality across both project mainlines. The main risks lie in screenshot extraction variability, response latency caused by retrieval and inference in the same request chain, and the sensitivity of interpersonal data, which makes privacy handling and permission control especially important throughout the project.

The document is intentionally written as a pre-execution project plan, even though the current repository already indicates technical exploration. All schedule language below therefore uses forward-looking, charter-style wording.

RBS

The Responsibility Breakdown Structure (RBS) defines the major responsibility areas of the HeartCompass course project. It clarifies the scope of each project stream, reduces overlap between related tasks, and ensures that both product mainlines remain balanced throughout execution.

User and Relationship Data Foundation

This responsibility stream will establish the identity and relationship baseline of the platform. It will cover user registration, profile management, MBTI collection, and the creation of Crush and RelationChain records so that every later function operates on a consistent interpersonal model. It will also define the ownership, validation, and maintenance rules that keep relationship data reliable across the project lifecycle.

Context Acquisition and Knowledge Management

This stream will convert fragmented interpersonal materials into structured context. Natural language narratives, screenshot materials, and curated knowledge entries will be normalized into reusable assets such as Event, ChatTopic, Knowledge, DerivedInsight, and ContextEmbedding records. The main responsibility here is to ensure that the stored context remains retrievable, well-typed, and suitable for downstream reasoning and recall.

Relationship Analysis Engine

This responsibility stream will organize the analytical core of HeartCompass. It will assemble context from profile facts, retrieved knowledge, relationship stage information, and interaction signals, then produce communication suggestions, risk prompts, and next-step guidance through narrative analysis and conversation analysis. The quality of this stream will be judged by relevance, clarity, and practical usefulness rather than by generic text generation alone.

Virtual Figure Interaction System

This stream will be responsible for the second project mainline, namely the virtual figure. It will load persona context, recall relevant relationship memories, and generate context-grounded dialogue through a structured graph workflow. It will also manage WebSocket interaction logic, timed message batching, and context refresh so that the virtual figure can remain coherent over multiple rounds of communication.

Client Experience, Integration, and Deployment

This stream will connect the user-facing interface with the backend intelligence modules and will ensure that the project can be demonstrated as a complete product rather than as isolated technical components. It will include client interaction design, full integration, system testing, performance optimization, and final packaging for course delivery.

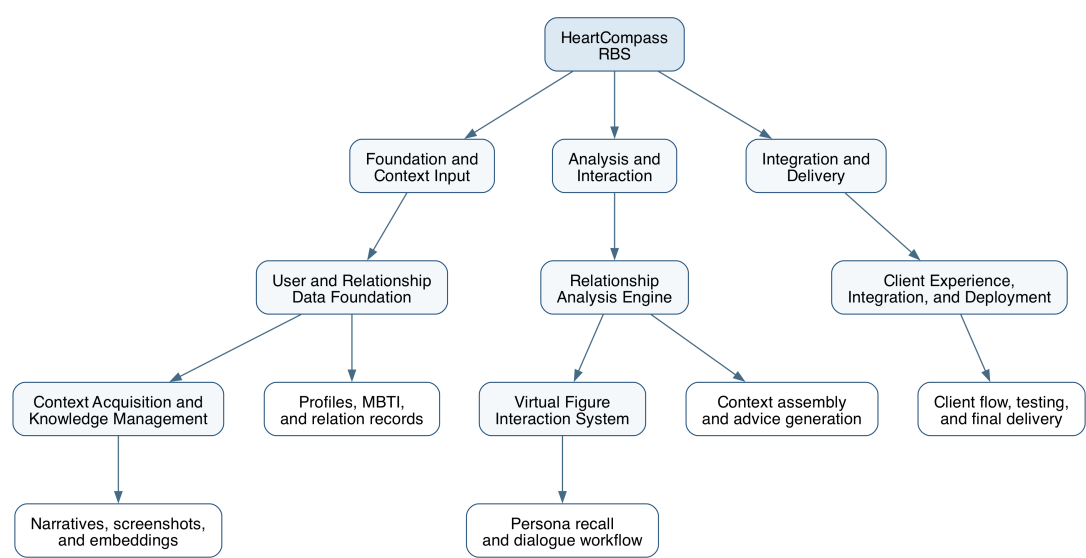


Figure 1. Responsibility Breakdown Structure for the HeartCompass course project.

WBS

The Work Breakdown Structure (WBS) translates the responsibility areas into manageable work packages, implementation activities, review checkpoints, and test targets. The project will estimate a total workload of 624 hours across all modules and will distribute the effort according to the table below.

The workload estimate places the greatest emphasis on the two core intelligence streams while reserving substantial effort for data foundation, integration, and delivery. This distribution is intended to keep the project academically credible as a software management exercise while still reflecting the real technical center of gravity of HeartCompass.

In practical terms, the WBS will serve as the bridge between the project charter and day-to-day execution. Each work package is defined not only by its functional goal, but also by its expected review focus and verification target. This helps the team translate strategic intent into manageable responsibilities and reduces the risk that major deliverables will be discussed at a high level without being converted into concrete work.

For management purposes, the following workload table summarizes the expected effort distribution across the five major work packages. The estimate is not intended as a rigid accounting device; rather, it provides a realistic planning baseline for balancing development depth, integration pressure,

and delivery obligations during the course period.

Work Package	Estimated Hours
User and Relationship Data Foundation	84
Context Acquisition and Knowledge Management	120
Relationship Analysis Engine	132
Virtual Figure Interaction System	126
Client Experience, Integration, and Deployment	162
Total	624

User and Relationship Data Foundation

The first work package will build the minimum data layer required by every other module. The team will implement user registration, login, profile management, MBTI collection, and the creation of Crush and RelationChain records with proper ownership and stage settings. During review, the team will verify that the data model supports both analysis and virtual figure scenarios without unnecessary duplication. Testing in this package will focus on CRUD reliability, permission isolation, and the consistency of relationship-state updates.

Context Acquisition and Knowledge Management

The second work package will handle the transformation of raw interpersonal materials into structured knowledge assets. The work will include natural language context extraction, screenshot-based topic extraction, knowledge entry management, and embedding generation. Review in this package will concentrate on data normalization quality and source consistency, while testing will verify extraction accuracy, embedding availability, and source-specific retrieval behavior.

Relationship Analysis Engine

The third work package will focus on constructing the analytical workflow. The team will implement context assembly, narrative analysis, conversation analysis, and the scoring logic that balances semantic relevance, importance, and temporal decay. Review activities will evaluate whether the generated suggestions are contextually meaningful and practically usable, and testing will cover narrative inputs, screenshot inputs, edge cases, and output stability under repeated runs.

Virtual Figure Interaction System

The fourth work package will implement the virtual figure mainline. It will cover persona loading, long-term memory recall, WebSocket communication rules, inactivity-triggered processing, and context_block refresh. Review will focus on whether the virtual figure preserves persona consistency and contextual faithfulness over several rounds of interaction. Testing will examine session continuity, response ordering, context refresh logic, and the behavior of extended chat scenarios.

Client Experience, Integration, and Deployment

The final work package will complete the product and prepare it for delivery. The team will develop a lightweight client for profile setup, context submission, result browsing, and virtual figure chat, then

integrate it with the backend APIs and graph workflows. Review at this stage will assess interface clarity, system coherence, and overall readiness for course presentation, while testing will include end-to-end acceptance checks, integration regression checks, performance observations, and final deployment validation.

Work Package Coordination

Although the work packages are listed separately, they will not be executed as isolated tasks. The data foundation and context acquisition packages will supply the core materials needed by both analytical and virtual-figure functions, while the client and deployment package will serve as the integration layer that brings all modules into a coherent demonstration flow. For this reason, the team will treat inter-package coordination as an explicit management activity rather than as an informal by-product of development.

From a scheduling perspective, the project will begin with planning and system design, then move into data and context construction, after which the analysis and virtual-figure streams will progress in parallel. Integration, testing, and optimization will follow only after the major technical branches have reached functional completeness. This arrangement is intended to reduce rework, keep dependency handling visible, and improve the likelihood of finishing the project within the assigned course window.

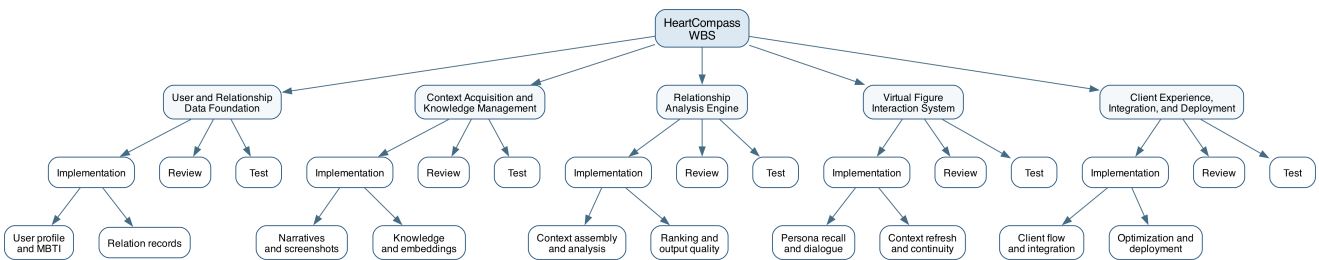


Figure 2. Work Breakdown Structure for the HeartCompass course project.

Gantt Chart

The project schedule is organized into nine coordinated phases. The timeline below presents the planned sequence, overlap, and delivery rhythm of the full product scope from requirements gathering to deployment.

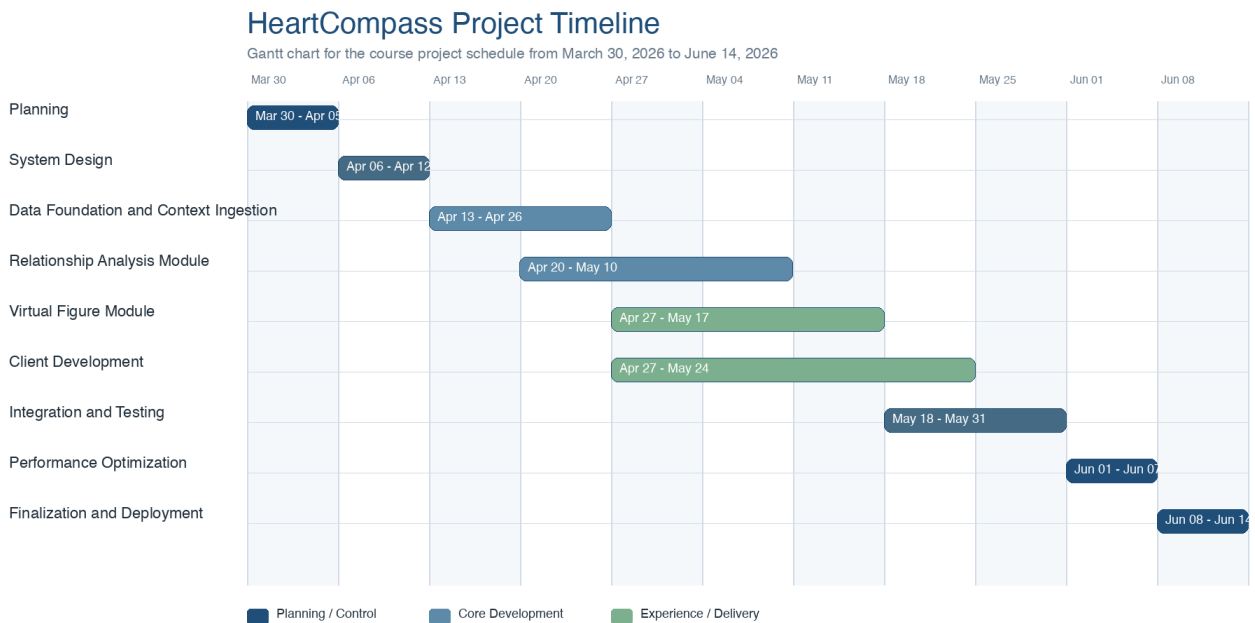


Figure 3. Gantt chart for HeartCompass from March 30, 2026 to June 14, 2026.

Planning (March 30, 2026 to April 05, 2026)

The team will define project objectives, scope, success criteria, and charter constraints so all later work can proceed on a stable foundation.

System Design (April 06, 2026 to April 12, 2026)

The system architecture, data flow, module interfaces, and technology choices will be finalized to support both analysis and virtual figure workflows.

Data Foundation and Context Ingestion (April 13, 2026 to April 26, 2026)

This phase will implement the user, relationship, and context ingestion layer that converts fragmented materials into structured business entities.

Relationship Analysis Module (April 20, 2026 to May 10, 2026)

The analysis pipeline will be developed to transform narrative and screenshot inputs into actionable communication suggestions and risk alerts.

Virtual Figure Module (April 27, 2026 to May 17, 2026)

The virtual figure workflow will be built to support context-grounded dialogue, persona loading, and memory-aware response generation.

Client Development (April 27, 2026 to May 24, 2026)

A lightweight user interface will be created for profile setup, material submission, result inspection, and virtual figure chat.

Integration and Testing (May 18, 2026 to May 31, 2026)

All modules will be integrated and tested to ensure that the full workflow behaves consistently across inputs, outputs, and interaction states.

Performance Optimization (June 01, 2026 to June 07, 2026)

Latency, retrieval efficiency, and interaction stability will be tuned so the platform remains usable under typical course project workloads.

Finalization and Deployment (June 08, 2026 to June 14, 2026)

The team will complete final bug fixing, documentation, packaging, and deployment to close the project on the planned final date.

Schedule Interpretation

Taken as a whole, the schedule shows a deliberate progression from definition to construction and from construction to integration. The project does not attempt to open all streams at once. Instead, it first stabilizes the planning logic and the system design, then uses the data and context layer as the shared foundation for the two principal functional streams. This arrangement is intended to keep major dependencies visible and to reduce the likelihood that later modules will be blocked by incomplete upstream work.

The overlap between the analysis stream, the virtual-figure stream, and client development is intentional rather than accidental. Once the shared foundation is in place, these workstreams can advance in parallel while still converging at integration and testing. In management terms, this is the stage at which coordination becomes most important, because quality problems in one stream can quickly propagate into the others if interface expectations are not kept aligned.

Critical Path Diagram

The critical path identifies the longest dependency chain that determines whether the project can finish on time. For HeartCompass, the critical path moves from requirements and architecture into data modeling, context ingestion, the virtual figure core, client integration, testing, optimization, and deployment. Relationship analysis is developed in parallel and converges during system integration.

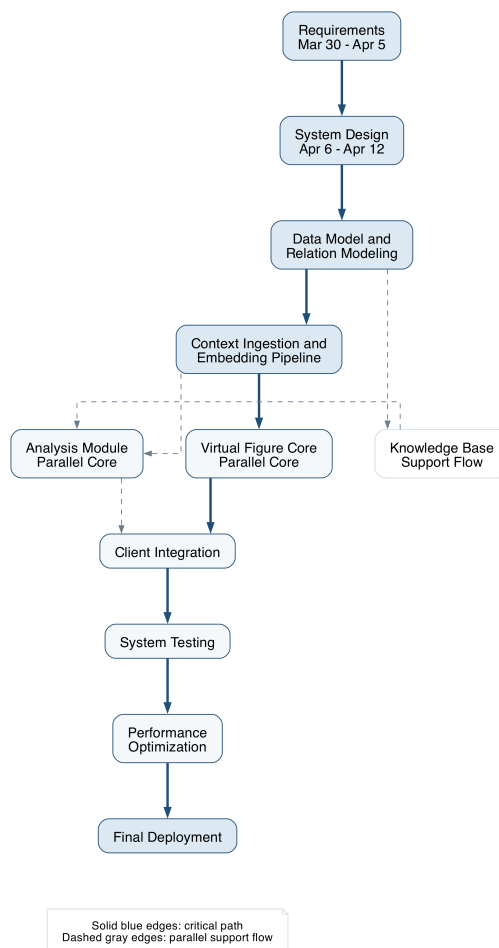


Figure 4. Critical path diagram highlighting the controlling dependency flow of the project.

Critical Path Interpretation

The project will begin with requirements gathering and system design because neither the data model nor the downstream AI modules can be stabilized before those decisions are made. Once the core relation model and context ingestion pipeline are defined, the virtual figure mainline becomes the controlling branch because it depends on context construction, memory design, and client integration in sequence. The relationship analysis module remains a major parallel branch; however, its outputs converge with the client flow during integration rather than controlling the overall finish date. This dependency structure makes system integration, test completion, and performance optimization the final schedule gates before deployment.

Planned Deliverables

By the end of the planned schedule, the team will deliver a documented data model, a backend service layer for context ingestion and retrieval, a relationship analysis workflow, a virtual figure interaction workflow, a lightweight client for demonstration, and a final integration package that includes testing evidence, documentation, and deployable output.

Project Governance Notes

Progress will be reviewed at the end of each major phase, and the schedule baseline, workload estimate, and critical path logic will serve as the main control instruments. Any scope adjustment will be evaluated against its likely effect on timeline, integration complexity, and delivery quality before it is accepted into the project plan.

Testing and Delivery Criteria

The final submission will need to satisfy several quality conditions at the same time. The charter must remain fully in English, use consistent future tense, and keep all dates aligned to 2026. Its sections, charts, workload estimate, and narrative explanations must remain mutually consistent, and the final PDF must be printable, readable, and professionally formatted without unstable spacing, clipped graphics, or broken page flow.